BAHRIA UNIVERSITY

Department of Software Engineering

LAB JOURNAL: 01

Introduction to Web Engineering

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Section: BSE-6A

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Repository: **Lab 01**

January 27, 2025

1 Lab Tasks

1.1 Discussion on Web Servers

A web server is software that receives HTTP/HTTPS requests from client browsers, processes them, and returns corresponding responses such as HTML pages, media files, or structured data. Modern web servers also handle security (TLS), access control, logging, compression, caching, and concurrent connections.

Different servers adopt different request-handling architectures. Process-based and thread-based designs allocate execution units per request, while event-driven servers use non-blocking I/O to efficiently serve large numbers of concurrent users.

Common web servers include:

- **Apache:** A widely-used open-source server known for modularity and flexibility. Supports multiple concurrency strategies and extensive configuration via .htaccess.
- **Nginx:** Event-driven and optimized for high concurrency and low memory usage. Frequently deployed as a reverse proxy, load balancer, or static content server.
- **IIS:** Microsoft's Windows-native server tightly integrated with ASP.NET, Active Directory, and enterprise security facilities.
- **LiteSpeed:** A high-performance commercial server with efficient PHP handling and drop-in compatibility with many Apache configurations.

Comparison Table of Common Web Servers

Web Server	Architecture / Design	Common Usage
Apache	Process/Thread-based, Modular	Shared hosting, .htaccess-heavy deployments
Nginx	Event-driven, Asynchronous	Reverse proxy, load balancing, static content
IIS	Windows-native, Thread-based	Enterprise .NET environments, Active Directory integration
LiteSpeed	Event-driven, PHP-optimized	High-traffic CMS (WordPress), commercial hosting

In practice, the choice of web server depends on performance requirements, hosting environment, stack compatibility, and administrative control.

1.2 Task 2: Demonstration of Basic HTML Concepts

The following HTML code demonstrates headings, italics, underlining, and hyperlinks as introduced in the lab session.

```
<html>
<head>
    <title> Introduction to Web Engineering</title>
</head>
<body>
    <p><h1> Talha Naveed </h1></p>
    <u><p><h3><i>BUIC STUDENT<br>
                Bahria School of Engineering and Applied Sciences (
                   BSEAS) <br> Department of Software Engineering (
                   BSE)
                <br></p></u>
    <a href="https://cms.bahria.edu.pk/Logins/Student/Login.aspx">
       University website</a> </i></h3>
</body>
</html>
```

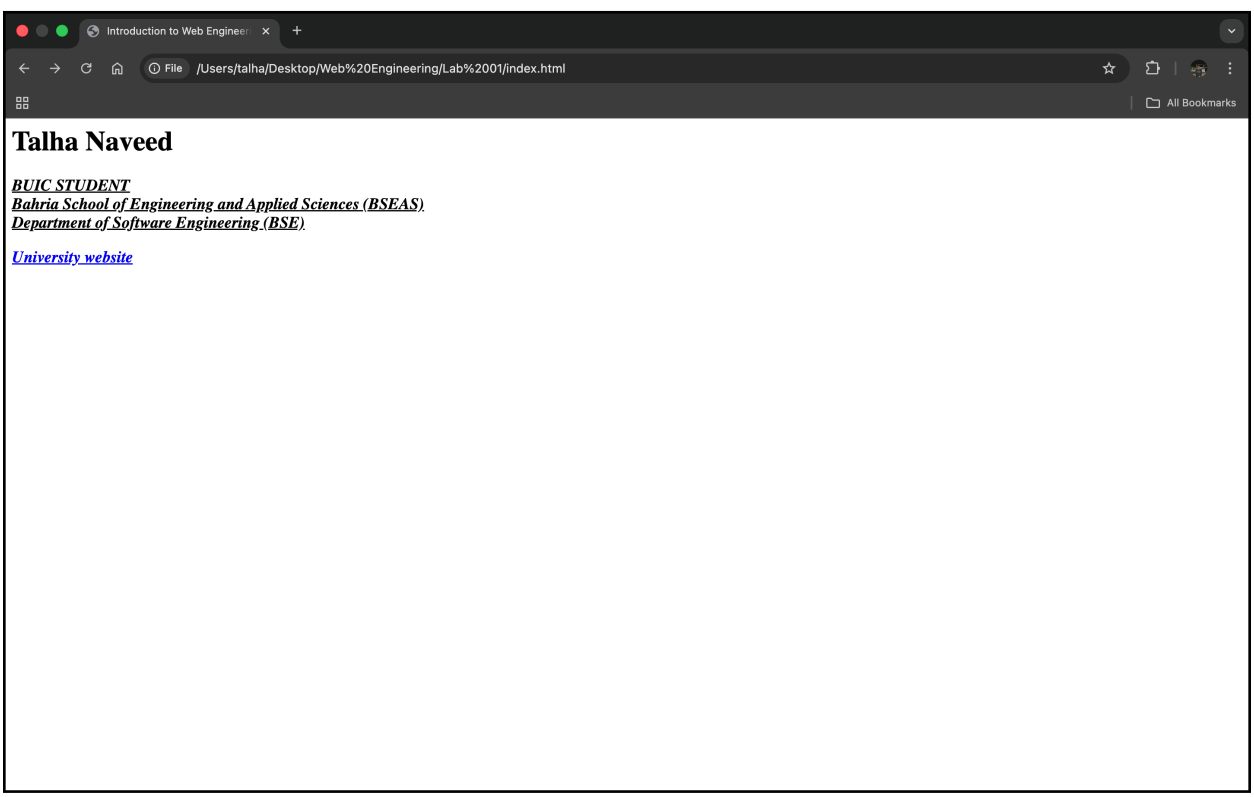
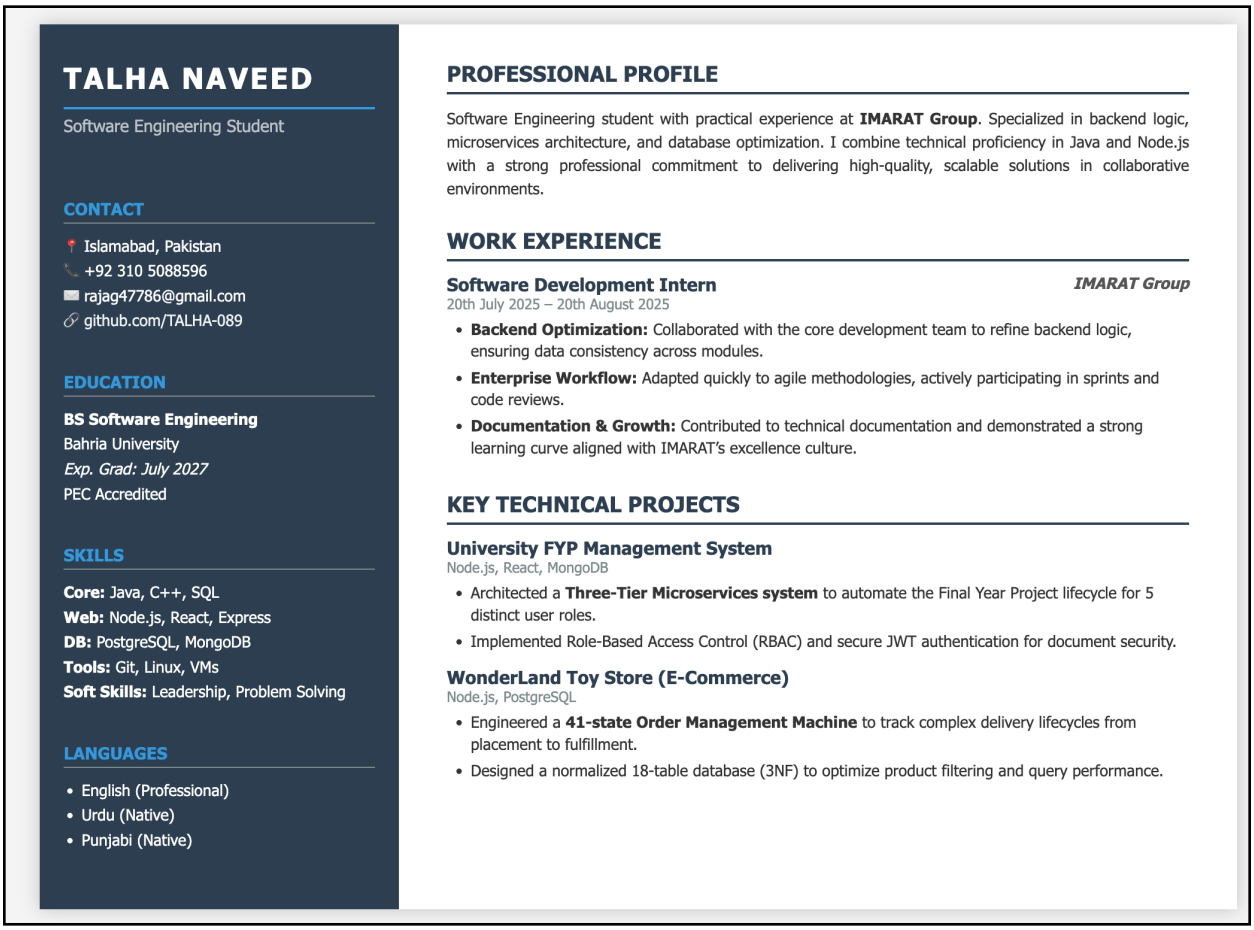


Figure 1.1: Output of Basic HTML Tags

1.3 Task 3: HTML CV Creation

A CV was developed using HTML to reinforce tags including headings, paragraphs, lists, formatting elements, and hyperlinks.



TALHA NAVEED

Software Engineering Student

CONTACT

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EDUCATION

BS Software Engineering
Bahria University
Exp. Grad: July 2027
PEC Accredited

SKILLS

Core: Java, C++, SQL
Web: Node.js, React, Express
DB: PostgreSQL, MongoDB
Tools: Git, Linux, VMs
Soft Skills: Leadership, Problem Solving

LANGUAGES

- English (Professional)
- Urdu (Native)
- Punjabi (Native)

PROFESSIONAL PROFILE

Software Engineering student with practical experience at **IMARAT Group**. Specialized in backend logic, microservices architecture, and database optimization. I combine technical proficiency in Java and Node.js with a strong professional commitment to delivering high-quality, scalable solutions in collaborative environments.

WORK EXPERIENCE

Software Development Intern — *IMARAT Group*
20th July 2025 – 20th August 2025

- **Backend Optimization:** Collaborated with the core development team to refine backend logic, ensuring data consistency across modules.
- **Enterprise Workflow:** Adapted quickly to agile methodologies, actively participating in sprints and code reviews.
- **Documentation & Growth:** Contributed to technical documentation and demonstrated a strong learning curve aligned with IMARAT's excellence culture.

KEY TECHNICAL PROJECTS

University FYP Management System
Node.js, React, MongoDB

- Architected a **Three-Tier Microservices system** to automate the Final Year Project lifecycle for 5 distinct user roles.
- Implemented Role-Based Access Control (RBAC) and secure JWT authentication for document security.

WonderLand Toy Store (E-Commerce)
Node.js, PostgreSQL

- Engineered a **41-state Order Management Machine** to track complex delivery lifecycles from placement to fulfillment.
- Designed a normalized 18-table database (3NF) to optimize product filtering and query performance.

Figure 1.2: HTML CV Output (with a little CSS)

2 Conclusion

In Lab 01, We focused on introductory concepts in Web Engineering, which provided hands-on exposure to basic HTML authoring. The exercises demonstrated the role of web servers, how requests are handled, and how simple markup can be rendered by browsers. These foundational tasks prepare us for more advanced topics in later labs.